

Qualification Pack



Blockchain Full Stack Developer

QP Code: SSC/Q8703

Version: 3.0

NSQF Level: 5

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Qualification Pack

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Qualification Pack

SSC/Q8703: Blockchain Full Stack Developer

Brief Job Description

Individuals at this job are responsible for developing the core blockchain. They will be responsible for developing tools, processes, and mechanisms for defining and developing the core blockchain. They will be responsible for defining and documenting technical specifications pertaining to the blockchain solution, including client, consensus algorithm, distributed data management, ledger maintenance, participant incentivization, smart contract/ chain code, and tokenomics.

Personal Attributes

Individuals in this job role must possess strong coordination and communication skills to collaborate with various stakeholders. They must also have a commitment to professional excellence and keep themselves abreast of the latest developments related to their field of work.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8738: Set up development environment for blockchain solution development](#)
2. [SSC/N8739: Develop backend services and frontend interfaces for the dApp, integrating all components.](#)
3. [SSC/N8740: Conduct comprehensive testing of the blockchain solution](#)
4. [SSC/N8741: Monitor the performance of blockchain applications and networks](#)
5. [SSC/N8728: Provide support in resolving issues related to blockchain apps](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Blockchain
Country	India
NSQF Level	5

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Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (of 3-year/ 4-year UG.PG/UG/Degree/Diploma in (Engg./Science)) OR Completed 3 year diploma after 10th (PG/UG/Degree/Diploma in (Engg./Science)) with 3 Years of experience in relevant field.Relevant experience in job roles related to Blockchain/Computer Science/IT The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (4) with 1.5 years of experience in relevant field.Relevant experience in job roles related to Blockchain/Computer Science/IT The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualifications.
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-05-IT-04331-2025-V2-NASSCOM
NQR Version	3.0

Remarks:

NA

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SSC/N8738: Set up development environment for blockchain solution development

Description

This NOS is about setting up the development environment for blockchain solution development

Scope

The scope covers the following :

- Set up development environment and implement smart contracts

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install Node.js and npm (Node Package Manager), these are essential for most blockchain development tools.
- PC2.** Choose an Integrated Development Environment (IDE) for example Visual Studio Code with Solidity extensions
- PC3.** Install Framework (for example truffle framework for ethereum)
- PC4.** Set up a local blockchain for development and testing
- PC5.** Install browser extension for wallets (eg metamask) to interact with the blockchain networks

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Familiarity with the core principles of blockchain, including decentralization, immutability, consensus mechanisms, and cryptographic security
- KU2.** Knowledge of different types of blockchains (public, private, consortium, hybrid) and their use cases.
- KU3.** Knowledge of programming languages like solidity for example ethereum smart contract development
- KU4.** Knowledge of development frameworks like truffle/Hardhat
- KU5.** Familiarity with tools like Mocha and Chai for contract testing
- KU6.** Knowledge of tools for running a local blockchain for development and testing
- KU7.** Knowledge of wallets to interact with the blockchain networks
- KU8.** Familiarity with Proof of Work, Proof of Stake, etc.
- KU9.** Knowledge of version control system like github

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** The ability to explain complex technical concepts to non-technical stakeholders is critical.
- GS2.** Effective communication fosters teamwork, enabling developers to work closely with designers, project managers, and other developers
- GS3.** Blockchain developers often face complex challenges that require strong analytical skills to evaluate potential solutions and make informed decisions.
- GS4.** The ability to assess the efficiency of different blockchain architectures and smart contract logic is essential for optimizing performance & security.
- GS5.** The ability to work collaboratively with various experts (e.g., engineers, business analysts) enhances innovation and project success.
- GS6.** Strong interpersonal skills help in mediating disagreements within the team, fostering a cooperative working environment
- GS7.** Managing multiple tasks and deadlines is essential in a project-driven environment. Developers must prioritize effectively to meet milestones.
- GS8.** Solid grasp of business objectives helps align blockchain solutions with organizational goals, ensuring that technology serves practical purposes.
- GS9.** The ability to manage time effectively ensures that projects stay on track and deadlines are met without compromising quality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Install Node.js and npm (Node Package Manager),these are essential for most blockchain development tools.	8	10	-	4
PC2. Choose an Integrated Development Environment (IDE) for example Visual Studio Code with Solidity extensions	6	10	-	4
PC3. Install Framework (for example truffle framework for ethereum)	7	10	-	4
PC4. Set up a local blockchain for development and testing	5	10	-	4
PC5. Install browser extension for wallets (eg metamask) to interact with the blockchain networks	4	10	-	4
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8738
NOS Name	Set up development environment for blockchain solution development
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8739: Develop backend services and frontend interfaces for the dApp, integrating all components.

Description

This NOS is about to Develop backend services and frontend interfaces for the dApp, integrating all components.

Scope

The scope covers the following :

- Planning and architecture
- Smart Contract Development
- Backend Development
- Front end development
- Testing, integration and deployment

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define the dApp's requirements and features
- PC2.** Choose the blockchain platform (e.g., Ethereum, Binance Smart Chain)
- PC3.** Design the system architecture
- PC4.** Plan the data flow between components
- PC5.** Develop Smart contracts
- PC6.** Set up a Node.js server (e.g., Express.js, Nest.js) for back end development
- PC7.** Implement RESTful API endpoints
- PC8.** Integrate Web3.js or Ethers.js for blockchain interactions
- PC9.** Set up a database (e.g., MongoDB, PostgreSQL)
- PC10.** Implement data synchronization between blockchain and database
- PC11.** Develop authentication and authorization mechanisms
- PC12.** Set up a React.js project (e.g., Create React App, Next.js) for frontend development
- PC13.** Design and implement UI components
- PC14.** Integrate Web3.js or Ethers.js for client-side blockchain interactions
- PC15.** Implement state management (e.g., Redux, Context API)
- PC16.** Create forms for user input and transaction submission
- PC17.** Develop pages for displaying blockchain data and transaction history
- PC18.** Develop comprehensive unit tests for smart contracts
- PC19.** Implement end-to-end testing strategies for dApps
- PC20.** Integrate smart contracts, backend services, and frontend interfaces
- PC21.** Deploy dApps to test networks and prepare for deployment

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Proficiency in HTML, CSS, and JavaScript for building web interfaces.
- KU2.** Expertise in React.js or other modern front-end frameworks like Angular or Vue.js.
- KU3.** Familiarity with Web3 libraries such as Web3.js or Ethers.js for blockchain interactions.
- KU4.** Knowledge of UI component libraries like Material-UI or Semantic UI for efficient interface design.
- KU5.** Understanding of state management solutions like Redux or MobX for complex application states.
- KU6.** Experience with wallet integration and blockchain transaction handling.
- KU7.** Proficiency in asynchronous programming and handling Promises in JavaScript.
- KU8.** Ability to integrate and work with RESTful APIs and potentially GraphQL.
- KU9.** Skills in unit testing and end-to-end testing for dApp front-ends.
- KU10.** Proficiency with version control systems, particularly Git and GitHub.
- KU11.** Familiarity with build tools and module bundlers like Webpack or Create React App.
- KU12.** Knowledge of front-end performance optimization techniques.
- KU13.** Understanding of web security best practices and blockchain-specific security considerations.
- KU14.** Basic comprehension of blockchain technology and the specific platform being used (e.g., Ethereum, Solana).
- KU15.** Grasp of UI/UX design principles for creating user-friendly interfaces.
- KU16.** Ability to implement responsive and mobile-first design for cross-device compatibility.
- KU17.** Knowledge of web accessibility standards and their implementation in dApps

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to design innovative and engaging user interfaces for blockchain applications.
- GS2.** Skill to tackle complex design and technical challenges in dApp development.
- GS3.** Flexibility to keep up with rapidly evolving blockchain technologies and design trends.
- GS4.** Capacity to clearly explain technical concepts to both technical and non-technical stakeholders.
- GS5.** Ability to work effectively with cross-functional teams, including backend developers and blockchain experts.
- GS6.** Meticulousness in creating pixel-perfect designs and spotting UI inconsistencies.
- GS7.** Skill to prioritize tasks and meet project deadlines in fast-paced development environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define the dApp's requirements and features	1	1	-	0.5
PC2. Choose the blockchain platform (e.g., Ethereum, Binance Smart Chain)	1	2	-	0.5
PC3. Design the system architecture	2	2	-	1
PC4. Plan the data flow between components	2	2	-	1
PC5. Develop Smart contracts	6	8	-	1
PC6. Set up a Node.js server (e.g., Express.js, Nest.js) for back end development	1	2	-	1
PC7. Implement RESTful API endpoints	1	2	-	1
PC8. Integrate Web3.js or Ethers.js for blockchain interactions	1	2	-	1
PC9. Set up a database (e.g., MongoDB, PostgreSQL)	1	2	-	1
PC10. Implement data synchronization between blockchain and database	1	2	-	1
PC11. Develop authentication and authorization mechanisms	1	2	-	1
PC12. Set up a React.js project (e.g., Create React App, Next.js) for frontend development	1	2	-	1
PC13. Design and implement UI components	1	2	-	1
PC14. Integrate Web3.js or Ethers.js for client-side blockchain interactions	1	2	-	1
PC15. Implement state management (e.g., Redux, Context API)	1	2	-	1
PC16. Create forms for user input and transaction submission	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Develop pages for displaying blockchain data and transaction history	1	2	-	1
PC18. Develop comprehensive unit tests for smart contracts	2	2	-	1
PC19. Implement end-to-end testing strategies for dApps	2	3	-	1
PC20. Integrate smart contracts, backend services, and frontend interfaces	1	3	-	1
PC21. Deploy dApps to test networks and prepare for deployment	1	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8739
NOS Name	Develop backend services and frontend interfaces for the dApp, integrating all components.
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SSC/N8740: Conduct comprehensive testing of the blockchain solution

Description

Conducting testing of a blockchain solution involves a systematic approach to ensure that the application functions correctly, securely, and efficiently

Scope

The scope covers the following :

- Define Test Objectives
- Test Planning
- Conduct different type of testing
- Monitoring and reporting

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Establish clear objectives for testing, such as verifying functionality, ensuring security, and assessing performance under various conditions.
- PC2.** Gather requirements from stakeholders to understand what needs to be tested, including specific functionalities and performance metrics.
- PC3.** Create detailed test cases covering all aspects of the application based on functional requirements.
- PC4.** Prepare a suitable testing environment that mimics production conditions as closely as possible
- PC5.** Check if the tests can be automated to improve efficiency and coverage.
- PC6.** Conduct functional testing to check transaction processing, smart contract execution, and adherence to consensus mechanisms.
- PC7.** Conduct integration testing to ensure that different components of the blockchain application (e.g., smart contracts, APIs) interact correctly.
- PC8.** Conduct performance testing to Assess the speed and efficiency of the blockchain network
- PC9.** Conduct security testing to identify vulnerabilities within the blockchain application.
- PC10.** Conduct Smart contracts testing to look for logical errors, proper execution paths, and potential security issues like reentrancy attacks or overflow errors
- PC11.** Conduct API testing to ensure that the APIs used for integrating with other software function correctly and securely
- PC12.** Record outcomes of each test case, including any failures or issues encountered.
- PC13.** Review results to identify patterns or recurring issues that need addressing.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Knowledge of how blockchain works, including its architecture, components (blocks, nodes, transactions), and types (public, private, consortium) is essential
- KU2.** Knowledge of functional testing to verify features of the blockchain application work as intended, including transaction processing and smart contract execution
- KU3.** Knowledge of integration testing to test interactions between different components of the blockchain application to ensure they work together seamlessly
- KU4.** Knowledge of performance testing to assess the speed and efficiency of the blockchain under various load conditions, measuring metrics like transaction throughput and latency.
- KU5.** Knowledge of security testing to identify vulnerabilities within the blockchain application, including testing for unauthorized access and resilience against attacks.
- KU6.** Understanding how to validate smart contracts to ensure they execute correctly according to predefined rules
- KU7.** Familiarity with Testing Tools and frameworks like truffle, gnache, populus etc.
- KU8.** Proficiency in languages like JavaScript or Python for writing test scripts and automating test cases.
- KU9.** Familiarity with Solidity (for Ethereum) or other relevant languages used in smart contract development.
- KU10.** Understanding how to test APIs that interact with blockchain applications, ensuring that they function correctly and securely.
- KU11.** Understanding cryptographic principles that secure transactions on the blockchain, including public/private key infrastructure and hashing algorithms.
- KU12.** Familiarity with best practices for securing blockchain applications against common vulnerabilities.
- KU13.** Ability to use performance monitoring tools to evaluate system performance under various conditions.
- KU14.** Knowledge of how to create detailed test cases that cover all aspects of the application based on functional requirements.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** The ability to analyze complex issues and identify the root causes of problems is crucial in testing blockchain applications
- GS2.** Effectively communicating technical concepts to non-technical stakeholders ensures that everyone involved understands the testing outcomes, potential issues, and implications for the project
- GS3.** Testing blockchain applications requires meticulous attention to detail to ensure that every aspect is thoroughly evaluated. Small errors can lead to significant vulnerabilities or functionality issues.
- GS4.** Blockchain projects often involve cross-functional teams. The ability to work collaboratively enhances innovation and ensures that all aspects of the application are considered during testing.
- GS5.** Building relationships with stakeholders allows testers to gather requirements effectively and understand the context in which the blockchain application will be used.

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- GS6.** Testers must manage multiple tasks and deadlines effectively. The ability to prioritize tasks based on urgency and importance helps maintain productivity in fast-paced environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Establish clear objectives for testing, such as verifying functionality, ensuring security, and assessing performance under various conditions.	3	5	-	1
PC2. Gather requirements from stakeholders to understand what needs to be tested, including specific functionalities and performance metrics.	3	5	-	1
PC3. Create detailed test cases covering all aspects of the application based on functional requirements.	3	5	-	1
PC4. Prepare a suitable testing environment that mimics production conditions as closely as possible	3	4	-	1
PC5. Check if the tests can be automated to improve efficiency and coverage.	2	4	-	1
PC6. Conduct functional testing to check transaction processing, smart contract execution, and adherence to consensus mechanisms.	2	4	-	1
PC7. Conduct integration testing to ensure that different components of the blockchain application (e.g., smart contracts, APIs) interact correctly.	2	4	-	2
PC8. Conduct performance testing to Assess the speed and efficiency of the blockchain network	2	4	-	2
PC9. Conduct security testing to identify vulnerabilities within the blockchain application.	2	3	-	2
PC10. Conduct Smart contracts testing to look for logical errors, proper execution paths, and potential security issues like reentrancy attacks or overflow errors	2	3	-	2
PC11. Conduct API testing to ensure that the APIs used for integrating with other software function correctly and securely	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Record outcomes of each test case, including any failures or issues encountered.	2	3	-	2
PC13. Review results to identify patterns or recurring issues that need addressing.	2	3	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8740
NOS Name	Conduct comprehensive testing of the blockchain solution
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SSC/N8741: Monitor the performance of blockchain applications and networks

Description

Performance optimization in blockchain development is crucial for enhancing transaction speed, scalability, and overall efficiency

Scope

The scope covers the following :

- Implement Layer 2 Solutions
- Hardware Optimization
- Network Optimization
- Performance Testing and Simulation
- Node level testing
- Smart Contract and DApp Performance
- Continuous Monitoring and Iteration

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Select appropriate consensus mechanism for optimizing transaction speed and energy consumption
- PC2.** Alleviate congestion on the main blockchain, by applying layer 2 solutions such as the Lightning Network etc. to be processed off-chain
- PC3.** Partition the blockchain into smaller segments (shards), enabling parallel processing of transactions
- PC4.** Ensure that nodes use solid-state drives (SSDs) for faster data access compared to traditional hard disk drives (HDDs)
- PC5.** Use CPUs with high clock speeds for better single-threaded performance, as blockchain nodes often benefit from fast processing capabilities rather than a high number of core
- PC6.** Optimize peer-to-peer (P2P) network topology to enhance communication between nodes
- PC7.** Implement selective verification strategies in consensus protocols to reduce communication overhead and improve transaction validation efficiency
- PC8.** Conduct performance testing using tools like Hyperledger Caliper to measure key metrics such as throughput and latency under various conditions
- PC9.** Use simulations to evaluate how changes in network configuration affect performance metrics, allowing for informed decisions regarding optimization strategies
- PC10.** Assess the resource utilization (CPU, memory, disk I/O) of blockchain nodes
- PC11.** Configure node exporters for collecting performance metrics & develop node performance dashboards
- PC12.** Optimize smart contracts for gas efficiency

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PC13. Implement monitoring tools to continuously track performance metrics, enabling proactive adjustments based on real-time data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of how blockchain systems are structured, including the roles of nodes, consensus mechanisms, and data layers.
- KU2.** Familiarity with various consensus algorithms (e.g., Proof of Work, Proof of Stake, Delegated Proof of Stake) and their impact on transaction speed and energy efficiency.
- KU3.** Knowledge of implementing Layer 2 solutions (e.g., state channels, sidechains) that allow off-chain transactions to enhance scalability and speed.
- KU4.** Knowledge of sharding techniques to partition the blockchain into smaller segments for parallel processing, improving throughput.
- KU5.** Knowledge of programming languages commonly used in blockchain development (e.g., Solidity for Ethereum smart contracts).
- KU6.** Familiarity with libraries like Web3.js or Ethers.js for integrating front-end applications with blockchain networks.
- KU7.** Knowledge of peer-to-peer network optimization techniques, such as gossip protocols and topology optimization, to enhance communication efficiency among nodes.
- KU8.** Knowledge of selective verification strategies to reduce communication overhead during transaction validation.
- KU9.** Awareness of hardware requirements for optimal node performance, including the use of SSDs for faster data access and CPUs with high clock speeds for efficient transaction processing.
- KU10.** Familiarity with tools like Hyperledger Caliper or custom scripts for benchmarking transaction throughput and latency under various conditions.
- KU11.** Ability to simulate different network configurations to assess their impact on performance metrics.
- KU12.** Knowledge of implementing monitoring solutions that track performance metrics continuously, allowing for proactive adjustments based on real-time data.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** The ability to critically analyze performance issues and devise effective solutions is crucial.
- GS2.** Analyzing metrics and performance data to make informed decisions about optimization strategies is essential for continuous improvement.
- GS3.** Effectively conveying complex technical concepts to non-technical stakeholders ensures that everyone involved understands the implications of performance optimizations.
- GS4.** Working closely with cross-functional teams (e.g., developers, project managers, and UX designers) requires clear communication to align objectives and share insights on performance improvements.

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- GS5.** Commitment to staying updated on the latest trends, techniques, and best practices in blockchain technology is vital for effective performance optimization
- GS6.** Managing multiple tasks and deadlines is crucial in a project-driven environment
- GS7.** Performance optimizations often require meticulous attention to detail to avoid introducing new issues while enhancing existing functionalities

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Select appropriate consensus mechanism for optimizing transaction speed and energy consumption	2	2	-	1
PC2. Alleviate congestion on the main blockchain, by applying layer 2 solutions such as the Lightning Network etc. to be processed off-chain	2	2	-	1
PC3. Partition the blockchain into smaller segments (shards), enabling parallel processing of transactions	2	2	-	1
PC4. Ensure that nodes use solid-state drives (SSDs) for faster data access compared to traditional hard disk drives (HDDs)	2	2	-	1
PC5. Use CPUs with high clock speeds for better single-threaded performance, as blockchain nodes often benefit from fast processing capabilities rather than a high number of core	2	2	-	1
PC6. Optimize peer-to-peer (P2P) network topology to enhance communication between nodes	2	5	-	1
PC7. Implement selective verification strategies in consensus protocols to reduce communication overhead and improve transaction validation efficiency	2	5	-	2
PC8. Conduct performance testing using tools like Hyperledger Caliper to measure key metrics such as throughput and latency under various conditions	2	5	-	2
PC9. Use simulations to evaluate how changes in network configuration affect performance metrics, allowing for informed decisions regarding optimization strategies	2	5	-	2
PC10. Assess the resource utilization (CPU, memory, disk I/O) of blockchain nodes	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Configure node exporters for collecting performance metrics & develop node performance dashboards	3	5	-	2
PC12. Optimize smart contracts for gas efficiency	3	5	-	2
PC13. Implement monitoring tools to continuously track performance metrics, enabling proactive adjustments based on real-time data.	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8741
NOS Name	Monitor the performance of blockchain applications and networks
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

SSC/N8728: Provide support in resolving issues related to blockchain apps

Description

This unit is about providing active support and troubleshooting issues in blockchain apps

Scope

The scope covers the following :

- Perform support-related activities

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define a plan for providing support, including third party services and internal resource allocation
- PC2.** Identify, analyze, and resolve complex technical issues within the blockchain ecosystem, providing timely support to maintain the stability and functionality
- PC3.** Determine the type of support required based on the application (Debugging, Scalability, Security enhancement, interoperability, CI/CD etc.)
- PC4.** Define the incident Management and Support Processes
- PC5.** Develop Continuous improvement and maintenance processes (analyzing user feedback, updated documentation, app optimization ec.)
- PC6.** Handle active support for blockchain end-users

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** types of blockchain
- KU2.** blockchain concepts such as nodes, consensus mechanism, smart contracts, forks, etc
- KU3.** concepts of hardware, networking, and data storage, especially those related to various systems in blockchain
- KU4.** ways to customize the network and apps in the current blockchain solution
- KU5.** concepts of blockchain security and cryptographic primitives such as keys, hash functions, encryption, signatures, etc.
- KU6.** processes related to node operations
- KU7.** processes related to chain management
- KU8.** on-chain governance rules
- KU9.** ways to implement enhancements/ upgrades in the blockchain network and apps
- KU10.** best practices and standard operating procedure associated with configuring/ customizing blockchain network and apps

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** ask for clarification and advice from appropriate people
- GS2.** conduct business contextual discussions
- GS3.** listen effectively and orally communicate information accurately
- GS4.** follow rule-based decision-making processes
- GS5.** make decisions on suitable courses
- GS6.** plan and organize the work to achieve targets and deadlines
- GS7.** work effectively in a customer facing environment
- GS8.** apply problem-solving approaches in different situations
- GS9.** apply balanced judgments to different situations
- GS10.** create accurate and well written documents/reports

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define a plan for providing support, including third party services and internal resource allocation	8	10	-	3
PC2. Identify, analyze, and resolve complex technical issues within the blockchain ecosystem, providing timely support to maintain the stability and functionality	6	12	-	3
PC3. Determine the type of support required based on the application (Debugging, Scalability, Security enhancement, interoperability, CI/CD etc.)	6	8	-	3
PC4. Define the incident Management and Support Processes	4	7	-	3
PC5. Develop Continuous improvement and maintenance processes (analyzing user feedback, updated documentation, app optimization ec.)	4	9	-	4
PC6. Handle active support for blockchain end-users	2	4	-	4
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8728
NOS Name	Provide support in resolving issues related to blockchain apps
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Blockchain
NSQF Level	5
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8738.Set up development environment for blockchain solution development	30	50	0	20	100	18
SSC/N8739.Develop backend services and frontend interfaces for the dApp, integrating all components.	30	50	0	20	100	18
SSC/N8740.Conduct comprehensive testing of the blockchain solution	30	50	0	20	100	18
SSC/N8741.Monitor the performance of blockchain applications and networks	30	50	0	20	100	18
SSC/N8728.Provide support in resolving issues related to blockchain apps	30	50	0	20	100	18
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	170	280	-	100	550	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.
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